# Variation and selection

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

## 1 ? Part 1

Core Variation [1]

**Answer: B**

**Explanation**

Variation describes differences between individuals of the same species, rather than differences between species or non-living materials.

## 2 ? Part 1

Core Variation [1]

**Answer: D**

**Explanation**

Body mass can take values throughout a range. The other examples have distinct categories without intermediate types.

## 3 ? Part 1

Core Variation [1]

**Answer: A**

**Explanation**

Discontinuous variation produces separate categories. Measurement precision and a continuous range do not define it.

## 4 ? Part 1

Core Variation [1]

**Answer: C**

**Explanation**

Different environments can change a continuous trait even in the same genotype. The observation does not exclude a genetic contribution in other comparisons.

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## 5 ? Part 1 Core Variation [1]

**Answer: D**

**Explanation**

Mutation is genetic change that can form a new allele. Acquired learning or training does not by itself create an inherited allele.

## 6 ? Part 1 Core Variation [1]

**Answer: B**

**Explanation**

Ionising radiation can damage DNA and increase mutation rate. The other changes are not generally recognised as mutagens.

## 7 ? Part 1 Supplement Variation [1]

**Answer: C**

**Explanation**

A gene mutation changes DNA bases randomly. Metabolism and physical changes alone are not changes to a gene's base sequence.

## 8 ? Part 1 Supplement Variation [1]

**Answer: A**

**Explanation**

Mutation creates new genetic changes and meiosis and random fertilisation generate new combinations. Acquired responses are not themselves genetic variation.

## 9 ? Part 1 Core Adaptive features [1]

**Answer: B**

**Explanation**

An adaptive feature is inherited and helps survival and reproduction in the relevant environment. Size alone or an acquired injury does not establish this.

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## 10 ? Part 1 Core Adaptive features [1]

**Answer: C**

**Explanation**

Avoiding daytime heat can reduce overheating and water loss. It does not remove nutritional needs or change the animal's trophic role.

## 11 ? Part 1 Supplement Adaptive features [1]

**Answer: D**

**Explanation**

A thick waxy cuticle reduces evaporation. A thin cuticle and open stomata tend to increase water loss; lack of storage is not a water-conserving adaptation.

## 12 ? Part 1 Core Selection [1]

**Answer: A**

**Explanation**

Reproductive success determines contribution to the next generation. Need does not create a useful allele and artificial selection is directed by humans.

## 13 ? Part 1 Core Selection [1]

**Answer: C**

**Explanation**

Selective breeding selects parents and offspring with desirable features over generations. Gene insertion is genetic modification, a different process.

## 14 ? Part 1 Supplement Selection [1]

**Answer: A**

**Explanation**

Treatment selects pre-existing or newly arising resistant variants. Resistance does not arise because bacteria choose or need it.

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## 15 ? Part 1 Supplement Selection [1]

**Answer: B**

**Explanation**

Artificial selection is directed by human choice. Both forms depend on heritable variation and can change populations over generations.

## 16 ? Part 1 Core Variation [4]

**Mark scheme**

**(a)** **[2]**

- The volunteers are individuals of the same species with different hand spans. (1)
- Hand span is continuous variation because it spans a range of values. (1)

**(b)** **[2]**

- Inherited genetic differences. (1)
- Environmental influences such as nutrition during growth. (1)

**Worked solution**

(a) The volunteers are individuals of the same species with different hand spans. Hand span is continuous variation because it spans a range of values.

(b) Inherited genetic differences. Environmental influences such as nutrition during growth.

## 17 ? Part 1 Core Variation [4]

**Mark scheme**

**(a)** **[2]**

- The phenotypes fall into a limited number of separate categories. (1)
- There are no intermediate ABO groups between those categories. (1)

**(b)** **[2]**

- Discontinuous variation is usually caused by genes alone. (1)
- Continuous variation is influenced by both genes and environment. (1)

**Worked solution**

(a) The phenotypes fall into a limited number of separate categories. There are no intermediate ABO groups between those categories.

(b) Discontinuous variation is usually caused by genes alone. Continuous variation is influenced by both genes and environment.

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## 18 ? Part 1 Core Variation [4]

**Mark scheme**

**(a)** **[2]**

- The range is 61 − 18 = 43 mm. (1)
- Leaf length shows continuous variation. (1)

**(b)** **[2]**

- Measure many leaves using the same measuring method. (1)
- Sample leaves at a similar developmental stage or position. (1)

**Worked solution**

(a) The range is 61 − 18 = 43 mm. Leaf length shows continuous variation.

(b) Measure many leaves using the same measuring method. Sample leaves at a similar developmental stage or position.

## 19 ? Part 1 Core Variation [4]

**Mark scheme**

**(a)** **[2]**

- Mutation is genetic change. (1)
- It can produce a new alternative form of a gene, an allele. (1)

**(b)** **[2]**

- Ionising radiation. (1)
- Some chemicals. (1)

**Worked solution**

(a) Mutation is genetic change. It can produce a new alternative form of a gene, an allele.

(b) Ionising radiation. Some chemicals.

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## 20 ? Part 1 Supplement Variation [4]

**Mark scheme**

**(a)** **[2]**

- It is a random change. (1)
- The change is in the base sequence of DNA. (1)

**(b)** **[2]**

- Meiosis produces genetically different gametes. (1)
- Random fertilisation combines gametes in different combinations. (1)

**Worked solution**

(a) It is a random change. The change is in the base sequence of DNA.

(b) Meiosis produces genetically different gametes. Random fertilisation combines gametes in different combinations.

## 21 ? Part 1 Core Variation [4]

**Mark scheme**

**(a)** **[2]**

- The seedlings began with the same genotype. (1)
- Different environments can affect a continuously varying feature such as mass. (1)

**(b)** **[2]**

- Keep water and mineral supply the same. (1)
- Compare plants of the same age under the same temperature. (1)

**Worked solution**

(a) The seedlings began with the same genotype. Different environments can affect a continuously varying feature such as mass.

(b) Keep water and mineral supply the same. Compare plants of the same age under the same temperature.

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## 22 ? Part 1 Core Variation [4]

**Mark scheme**

**(a)** **[2]**

- Body length can have intermediate values across a range. (1)
- Seed shape has separate categories without intermediates. (1)

**(b)** **[2]**

- Measure lengths with a ruler using consistent units. (1)
- Count seeds in the smooth and wrinkled categories rather than measuring an intermediate shape score. (1)

**Worked solution**

(a) Body length can have intermediate values across a range. Seed shape has separate categories without intermediates.

(b) Measure lengths with a ruler using consistent units. Count seeds in the smooth and wrinkled categories rather than measuring an intermediate shape score.

## 23 ? Part 1 Supplement Variation [4]

**Mark scheme**

**(a)** **[2]**

- A gene mutation can create it. (1)
- A gene mutation is a random change in the DNA base sequence. (1)

**(b)** **[2]**

- Meiosis produces gametes with different genetic combinations. (1)
- Random mating or random fertilisation brings together different existing alleles. (1)

**Worked solution**

(a) A gene mutation can create it. A gene mutation is a random change in the DNA base sequence.

(b) Meiosis produces gametes with different genetic combinations. Random mating or random fertilisation brings together different existing alleles.

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## 24 ? Part 1 Supplement Adaptive features [4]

**Mark scheme**

**(a)** **[2]**

- Rolled leaves trap moist air and reduce air movement near stomata. (1)
- Sunken stomata retain humid air, reducing the water-vapour concentration gradient. (1)

**(b)** **[2]**

- A thick cuticle reduces evaporation from the leaf surface. (1)
- Storage tissue retains water for use when water uptake is limited. (1)

**Worked solution**

(a) Rolled leaves trap moist air and reduce air movement near stomata. Sunken stomata retain humid air, reducing the water-vapour concentration gradient.

(b) A thick cuticle reduces evaporation from the leaf surface. Storage tissue retains water for use when water uptake is limited.

## 25 ? Part 1 Core Adaptive features [4]

**Mark scheme**

**(a)** **[2]**

- Webbed feet provide a larger surface for propulsion through water. (1)
- Waterproof feathers help keep the body dry and insulated. (1)

**(b)** **[2]**

- It must be inherited. (1)
- It must help survival and reproduction in that environment. (1)

**Worked solution**

(a) Webbed feet provide a larger surface for propulsion through water. Waterproof feathers help keep the body dry and insulated.

(b) It must be inherited. It must help survival and reproduction in that environment.

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## 26 ? Part 1 Core Selection [4]

**Mark scheme**

**(a)** **[2]**

- Predators are less likely to detect and eat dark beetles. (1)
- More dark beetles survive to reproduce. (1)

**(b)** **[2]**

- Surviving dark beetles pass their alleles to offspring. (1)
- The alleles producing the advantageous colour become more common. (1)

**Worked solution**

(a) Predators are less likely to detect and eat dark beetles. More dark beetles survive to reproduce.

(b) Surviving dark beetles pass their alleles to offspring. The alleles producing the advantageous colour become more common.

## 27 ? Part 1 Core Selection [4]

**Mark scheme**

**(a)** **[2]**

- Select parents with high milk yield or a family history of high yield. (1)
- Cross the selected individuals. (1)

**(b)** **[2]**

- Select offspring that show high milk yield. (1)
- Breed the selected offspring and repeat over many generations. (1)

**Worked solution**

(a) Select parents with high milk yield or a family history of high yield. Cross the selected individuals.

(b) Select offspring that show high milk yield. Breed the selected offspring and repeat over many generations.

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## 28 ? Part 1 Supplement Selection [4]

**Mark scheme**

**(a)** **[2]**

- Susceptible bacteria are killed or their reproduction is inhibited. (1)
- Resistant bacteria survive more successfully. (1)

**(b)** **[2]**

- Surviving resistant bacteria reproduce. (1)
- Their offspring inherit resistance, increasing its frequency in the population. (1)

**Worked solution**

(a) Susceptible bacteria are killed or their reproduction is inhibited. Resistant bacteria survive more successfully.

(b) Surviving resistant bacteria reproduce. Their offspring inherit resistance, increasing its frequency in the population.

## 29 ? Part 1 Supplement Selection [4]

**Mark scheme**

**(a)** **[2]**

- Inherited drought-tolerant variants survive and reproduce more successfully. (1)
- Their alleles become more common over many generations. (1)

**(b)** **[2]**

- Environmental conditions determine which wild variants reproduce most successfully. (1)
- The farmer chooses which cultivated plants reproduce for seed yield. (1)

**Worked solution**

(a) Inherited drought-tolerant variants survive and reproduce more successfully. Their alleles become more common over many generations.

(b) Environmental conditions determine which wild variants reproduce most successfully. The farmer chooses which cultivated plants reproduce for seed yield.

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## 30 ? Part 1 Core Selection [4]

**Mark scheme**

**(a)** **[2]**

- Food and other resources are limited. (1)
- Many young compete for those resources and some die. (1)

**(b)** **[2]**

- Individuals already differ in inherited features. (1)
- Those with advantageous features leave more offspring without choosing to change their alleles. (1)

**Worked solution**

(a) Food and other resources are limited. Many young compete for those resources and some die.

(b) Individuals already differ in inherited features. Those with advantageous features leave more offspring without choosing to change their alleles.

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